



X-ray imaging from nonlinear waves

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Joint work with

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PICOF Inverse Problems, Control, and Shape Optimization 2025,
Hammamet, Tunis

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Problem setup

Let $\Omega \subset \mathbb{R}^n$ be a Lipschitz domain.

We are interested in the recovery of the potential function q in the semilinear wave equation

$$\begin{cases} \square u(t, x) + q(t, x)u(t, x)^m = 0, & \text{in } [0, T] \times \Omega, \\ u = f, & \text{on } [0, T] \times \partial\Omega, \\ u(0, x) = \partial_t u(0, x) = 0, & \text{on } \Omega, \end{cases}$$

Here $\square = \partial_t^2 - \Delta$ and we assume that the exponent m is an integer ≥ 2 .

The semilinear wave equation admits a unique solution

$$u \in \bigcap_{k=0}^s C^k(0, T; H^{s-k}(\Omega))$$

when $f \in H^s([0, T] \times \partial\Omega)$, $s > n/2$, is sufficiently small.

Notation and setting

$$\begin{cases} \square u(t, x) + q(t, x)u(t, x)^m = 0, & \text{in } [0, T] \times \Omega, \\ u = f, & \text{on } [0, T] \times \partial\Omega, \\ u(0, x) = \partial_t u(0, x) = 0, & \text{on } \Omega, \end{cases}$$

We will let

$$\Sigma = [0, T] \times \partial\Omega$$

denote the lateral boundary of the domain.

The Dirichlet-to-Neumann map (DN map) is defined by

$$\begin{aligned} \Lambda : H_c^s(\Sigma) &\rightarrow H^{s-1}(\Sigma), \\ \Lambda(f) &= \partial_\nu u_f|_\Sigma, \end{aligned}$$

where ν is the outward normal vector to Σ .

The inverse problem

$$\begin{cases} \square u(t, x) + q(t, x)u(t, x)^m = 0, & \text{in } [0, T] \times \Omega, \\ u = f, & \text{on } [0, T] \times \partial\Omega, \\ u(0, x) = \partial_t u(0, x) = 0, & \text{on } \Omega, \end{cases}$$

Inverse Problems

Reconstruct $q(x, t)$ from the DN map Λ . How stable is this reconstruction process?

Some earlier results

- ▶ Uniqueness for an inverse problem for $(\partial_t^2 - c(x)^2 \Delta)u = 0$ by boundary control methods by Belishev, Belishev-Kurylev in 1987-1995
- ▶ First results exploiting nonlinearity of the equation by Kurylev, Lassas and Uhlmann from 2014
- ▶ Since then, techniques using nonlinearity as a tool have been extremely popular: T Balehowsky, C Cârstea, X Chen, M de Hoop, A Feizmohammadi, C Guillarmou, P Hintz, D Johansson, Y Kian, H Koch, K Krupchyk, M Lassas, T Liimatainen, Y-H Lin, G Nakamura, L Oksanen, G Paternain, A Rüland, M Salo, P Stefanov, G Uhlmann, Y Wang, J Zhai, and many more (apologies to any who I missed!)

Assume that $q_1, q_2 \in C^{s+1}(\mathbb{R} \times \Omega)$ satisfy $\|q_j\|_{C^{s+1}} \leq c$, $j = 1, 2$, for some $c > 0$ and $s > (n-1)/2$. Let $\Lambda_1, \Lambda_2 : H_c^{s+1}(\Sigma) \rightarrow H^r(\Sigma)$, $r \leq s$, be the corresponding Dirichlet-to-Neumann maps of the non-linear wave equation.

Let $\varepsilon_0 > 0$ and $L > 0$. Let $\delta \in (0, L)$ be such that

$$\|\Lambda_1(f) - \Lambda_2(f)\|_{H^r(\Sigma)} \leq \delta$$

for all $f \in H_0^{s+1}(\Sigma)$ with $\|f\|_{H^{s+1}(\Sigma)} \leq \varepsilon_0$. Then there exists a constant $C > 0$, independent of q_1, q_2 , and $\delta > 0$, such that

$$\|q_1 - q_2\|_{L^\infty(W)} \leq C\delta^{\sigma(s,m)},$$

where

$$\sigma(s, m) = \frac{8(m-1)}{2m(m-1)(8s-n+13) + 2m-1}.$$

Ideas of the proof: higher-order linearization

Use a small parameter $\varepsilon > 0$:

$$\begin{cases} \square u(t, x) + q(t, x)u(t, x)^m = 0, & \text{in } [0, T] \times \Omega, \\ u = \varepsilon f, & \text{on } [0, T] \times \partial\Omega, \\ u(0, x) = \partial_t u(0, x) = 0, & \text{on } \Omega. \end{cases}$$

Then, due to the **nonlinearity**, by Taylor expanding around $\varepsilon = 0$ we find

$$u_{\varepsilon f} = 0 + \varepsilon v + \varepsilon^m w + \mathcal{R}(\varepsilon^{m+1}).$$

Here the function v satisfies

$$\begin{cases} \square v = 0, & \text{in } [0, T] \times \Omega, \\ v = f, & \text{on } \Sigma \\ v = \partial_t v = 0, & \text{at } t = 0 \end{cases}$$

Ideas of the proof: higher-order linearization

Moreover, the m th order term $w = D_\varepsilon^m u|_{\varepsilon=0}$ satisfies

$$\begin{cases} \square w + m! q v^m = 0, & \text{in } [0, T] \times \Omega, \\ w = 0, & \text{on } \Sigma \\ w = \partial_t w = 0, & \text{at } t = 0 \end{cases}$$

We produced a linear equation with a **controllable source**!

An integral identity

Integrating the differentiated equation against any solution v_0 of $\square v_0 = 0$ that vanishes at $T = 0$, we get

$$\begin{aligned}
 -m! \int_{[0, T] \times \Omega} \textcolor{red}{q} v_0 v^m dx dt &= \int_{[0, T] \times \Omega} v_0 \square w dx dt \\
 &= \int_{\Sigma} v_0 \partial_{\nu} w dS \\
 &= \int_{\Sigma} v_0 D_{\varepsilon}^m \Lambda(\varepsilon f) \Big|_{\varepsilon=0} dS
 \end{aligned}$$

An integral identity

Integrating the differentiated equation against v_0 , we get the identity

$$-m! \int_{[0, T] \times \Omega} \textcolor{red}{q} v_0 v^m dx dt = \int_{\Sigma} v_0 D_{\varepsilon}^m \Lambda(\varepsilon f) \Big|_{\varepsilon=0} dS$$

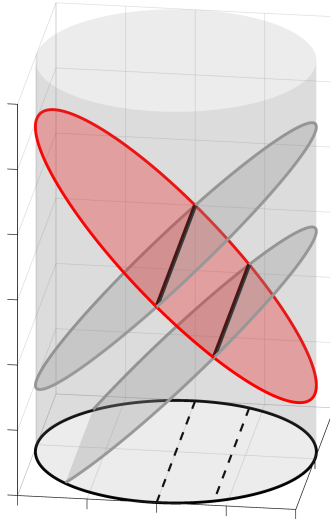
Idea: choose a family of (linear) waves, such that

$$v_0 v^m \approx \delta(x \cdot \theta = \eta) \delta(t - t_0)$$

Then

$$-m! \int_{[0, T] \times \Omega} \textcolor{red}{q} v_0 v^m dx dt \approx \int_{x \cdot \theta = \eta} q(x, t_0) dx = \mathcal{R}(q)(t_0, \theta, \eta).$$

Radon projections from waves



Choice of v_0 and v

Goal: $v_0 v^m \approx \delta((x \cdot \theta - \eta)\delta(t - t_0))$.

Define first

$$H(\ell) = \varphi_h(\ell) \tau^{1/2} e^{-\frac{1}{2}\tau \ell^2},$$

where $\varphi_h \in C_0^\infty(\mathbb{R}^n)$ with $\varphi_h(0) = 1$ (I will omit this function for the sake of clarity.)

Then we can simply choose

$$v_0(x, t) = \frac{1}{\pi} H(x \cdot \theta - t - (\eta_0 - t_0))$$

and

$$v(x, t) = \sqrt[m]{H(x \cdot \theta - t + (\eta_0 - t_0))}$$

to obtain

$$v_0 v^m = \frac{\tau}{\pi} e^{-\tau((x \cdot \theta - \eta_0)^2 + (t - t_0)^2)}$$

Approximation property

We have

$$v_0 v^m = \frac{\tau}{\pi} e^{-\tau((x \cdot \theta - \eta_0)^2 + (t - t_0)^2)}.$$

Let $t, \eta \in \mathbb{R}$ and $\theta \in \mathbb{S}^{n-1}$ and define

$$\mathcal{R}(q)(t, \theta, \eta) := \int_{x \cdot \theta = \eta} q(x, t) dS_x.$$

Lemma

If $q \in C^1(\mathbb{R}^n \times \mathbb{R})$, then there is $C > 0$ such that

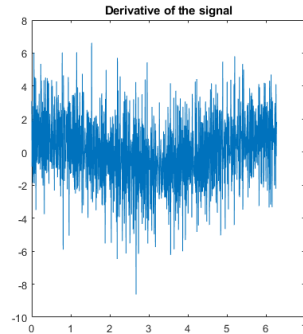
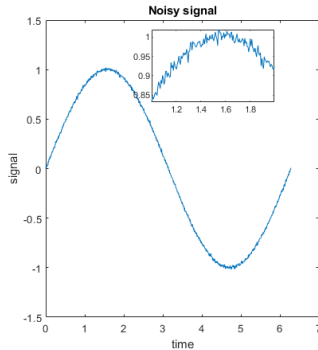
$$\left| \mathcal{R}(q)(t_0, \theta, \eta) - \frac{\tau}{\pi} \int_{\mathbb{R}^n \times \mathbb{R}} q(x, t) e^{-\tau((x \cdot \theta - \eta)^2 + (t - t_0)^2)} dx dt \right| \leq \frac{C}{\sqrt{\tau}}.$$

Minor hiccup

Recall we need

$$\int_{\Sigma} v_0 D_{\varepsilon}^m \Lambda(\varepsilon f) dS_x$$

Noisy measurement of $\Lambda(f) \implies$ derivatives blow up



Regularized higher-order differentiation

Let $g \in H^s(\mathbb{R})$ and suppose $g_\delta \in L^2(\mathbb{R})$, $\delta > 0$ with

$$\|g - g_\delta\|_{L^2} < \delta.$$

The exact derivative satisfies

$$g^{(p)}(x) = \mathcal{F}^{-1}((i\xi)^p \widehat{g}(\xi))(x).$$

Any high-frequency noise blows up, so we low-pass filter:

$$R_\alpha^{(p)}(g)(x) = \mathcal{F}^{-1}((i\xi)^p e^{-\alpha\xi^2} \widehat{g}(\xi))(x).$$

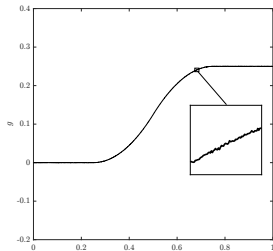
Theorem

Let $0 \leq p \leq s \leq p+2$ and $\alpha = \delta^{s/2}$. Then there exists $C > 0$ such that

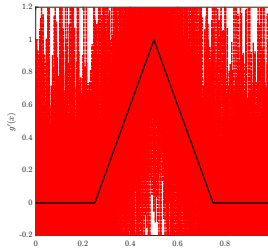
$$\|g^{(p)} - R_\alpha^{(p)}(g_\delta)\|_{L^2} \leq C(1 + \|g\|_{H^s})\delta^{1-p/s}.$$

Noise blows up...

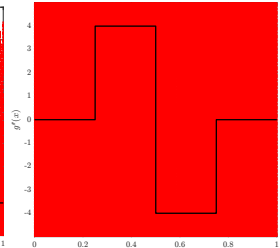
Noisy signal



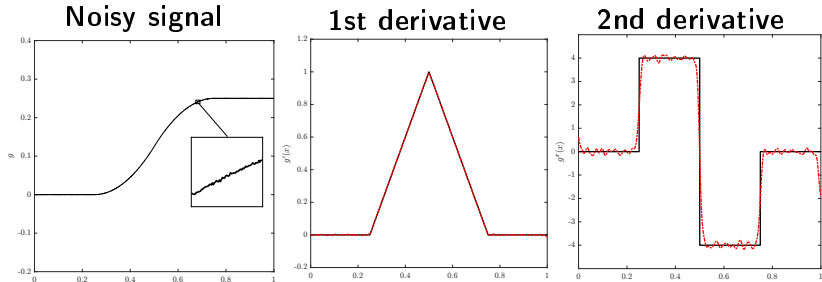
1st derivative



2nd derivative



...but regularization helps



Towards implementation: Reconstruction formula

The final reconstruction formula is then

$$\mathcal{R}^{\text{rec}}(q)(t_0, \theta, \eta) = R_{\alpha}^{(m)} \int_{\Sigma} v_0 \Lambda(\varepsilon f) dS$$

where $R_{\alpha}^{(m)}$ differentiates wrt. ε at $\varepsilon = 0$, and the parameters (t_0, θ, η) are hidden in the boundary values f .

- ▶ $R_{\alpha}^{(m)}$ can be implemented via FFT
- ▶ The Radon (X-ray) transform is inverted using standard filtered backprojection with Ram-Lak filter, i.e.,

$$f(x) = C \int_0^{\pi} (\mathcal{R}(f) * h)(x \cdot n_{\theta}) d\theta$$

$$\text{s.t. } \hat{h} = |\xi|.$$

Numerical examples

We use standard finite difference scheme for the forward problem on the square

$$\Omega = [-0.5, 0.5] \times [-0.5, 0.5]$$

over the time-interval $[0, 3]$. DN-map is measured with 32 different values of $\varepsilon \in [0, \varepsilon_0]$.

- ▶ For the plane waves: $\tau = 700$
- ▶ Regularization: $\alpha = 0.01$.
- ▶ Reconstruction over

$$\Theta = \{1^\circ, \dots, 179^\circ\}$$

and 63 uniformly distributed distances $\eta \in [-0.4, 0.4]$.

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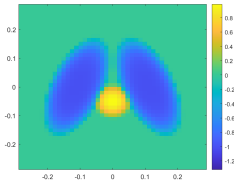
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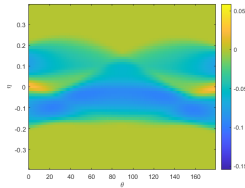
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Example of reconstructed sinogram

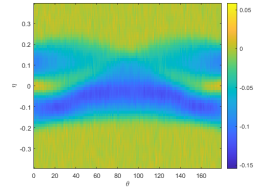
Potential q



True sinogram



Reconstruction

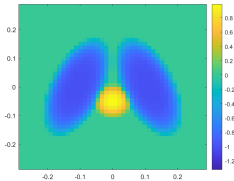


- Noise level: 2% (SNR: 40 dB)

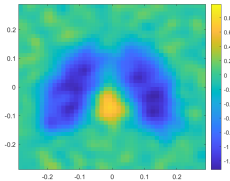
Pictures by S. Anttila

Example 1: Lung phantom

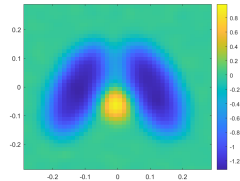
True potential q



No regularization



Regularized

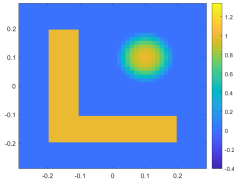


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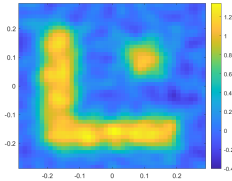
Pictures by S. Anttila

Example 2: Discontinuity

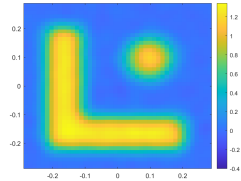
True potential q



No regularization



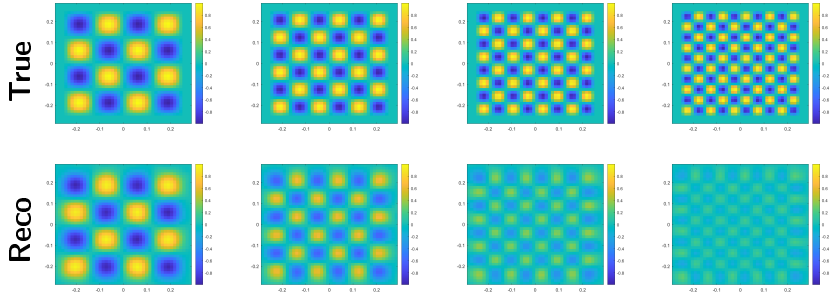
Regularized



► Noise level: 2% (SNR: 40 dB)

Pictures by S. Anttila

Example 3: Checkerboard



► Noise level: 2% (SNR: 40 dB)

Pictures by S. Anttila

Conclusion

$$\begin{cases} \square u(t, x) + q(t, x)u(t, x)^m = 0, & \text{in } [0, T] \times \Omega, \\ u = f, & \text{on } [0, T] \times \partial\Omega, \\ u(0, x) = \partial_t u(0, x) = 0, & \text{on } \Omega, \end{cases}$$

We have

- ▶ stability estimates (Hölder type)
- ▶ reconstruction formula
- ▶ regularization method and justified choice of regularization parameter
- ▶ non-iterative inversion: fast implementation through FFT based methods.

Thank you!



References

- ▶ Harju M, Tyni T, Takalahti S (now Anttila), *X-ray imaging from nonlinear waves: numerical reconstruction of a cubic nonlinearity*, arXiv:2509.11951, 2025.
- ▶ Lassas M, Liimatainen T, Potenciano-Machado L, and Tyni T, An inverse problem for a semi-linear wave equation: a numerical study, *Inv. Prob. & Imag.*, DOI: 10.3934/ipi.2023022, 2023
- ▶ Lassas M, Liimatainen T, Potenciano-Machado L, and Tyni T, Uniqueness, reconstruction and stability for an inverse problem of a semi-linear wave equation, *J. Diff. Eq.*, DOI:10.1016/j.jde.2022.08.010, 2022